

IT SUPPORT TROUBLESHOOTING LOG

Ubuntu Innovations (Pty) Ltd
Week 1 | May 2026

Organisation	Ubuntu Innovations (Pty) Ltd
Location	Cape Town, South Africa
Prepared by	[Your Full Name]
Programme	Capaciti Tech Career Accelerator
Certificate	Google IT Support Certificate
Report Period	Week 1 — May 2026
Total Tickets	3 (INC-001, INC-002, INC-003)
All Resolved	Yes — within 40 minutes each

1. Introduction

This report documents the IT support incidents logged and resolved during Week 1 (May 2026) at Ubuntu Innovations (Pty) Ltd. It was prepared as part of the Capaciti Tech Career Accelerator programme in fulfilment of the Google IT Support Certificate practicum requirements.

Three support tickets were raised and resolved within the reporting period, covering hardware, network, and peripheral issues across three departments. Each incident is documented with its reported symptoms, possible causes, troubleshooting methodology, root cause analysis, resolution, and recommended preventive measures.

2. IT Environment Overview

The following table summarises the infrastructure supporting all 25 users across 5 departments at Ubuntu Innovations.

Users	25 (across 5 departments)
Operating System	Windows 11 Pro (all workstations)
Network Topology	Star topology — 24-port managed switch

Router / Firewall	FortiGate 60F
File Server / NAS	Synology NAS — 192.168.10.10
Network Design	VLAN-segmented (4 VLANs)
Endpoint Security	Kaspersky Endpoint Security (24-hr definition updates)
Power Protection	APC Back-UPS Pro 1500VA on critical systems

2.1 VLAN Reference

The network is segmented into four VLANs to isolate traffic between device categories:

VLAN	Name	Subnet	Assigned Devices
VLAN 10	Servers	192.168.10.0/24	NAS (192.168.10.10), future servers
VLAN 20	User Devices	192.168.20.0/24	Desktops (.10–.29), Laptops (.30–.34)
VLAN 30	Printers	192.168.30.0/24	Printer 1 (.10), Printer 2 (.11)
VLAN 40	Wi-Fi	192.168.40.0/24	AP1 (.2), AP2 (.3)
—	Gateway	192.168.1.1	Router / FortiGate 60F Firewall
—	Switch Mgmt	192.168.1.2	24-port Managed Switch

3. Incident Summary

The following table provides an at-a-glance view of all support tickets raised during Week 1:

Ticket	Department	Issue	Resolution Time	Status
INC-001	Finance	Desktop not powering on	~25 minutes	Resolved
INC-002	Sales & Marketing	No internet connectivity	~40 minutes	Resolved
INC-003	Human Resources	Printer offline	~30 minutes	Resolved

4. Incident Reports

4.1 INC-001 — Desktop Not Powering On

Date Reported	19 May 2026
Reported By	Finance Department (Workstation: 192.168.20.12)

Assigned To	IT Support Specialist
Priority	High
Status	Resolved
Resolution Time	~25 minutes

Problem Description

A Finance department employee reported that their desktop computer (Intel Core i5 12th Gen, 16 GB RAM, 512 GB SSD, Windows 11 Pro) would not turn on when the power button was pressed. The screen remained black with no fan noise, LED indicators, or POST beeps. The workstation is used for Sage Business Cloud Accounting and Microsoft 365 and is assigned to VLAN 20.

Symptoms

- No lights, fans, or sounds when the power button was pressed.
- Monitor displayed "No Signal" and entered sleep mode.
- Computer was functioning normally the previous business day.
- UPS unit connected to the workstation showed a solid green indicator, suggesting power was available at the UPS output.

Possible Causes

- Loose or unplugged power cable at the wall socket or PSU inlet.
- Faulty or tripped power strip / wall socket.
- Blown fuse in the South African 3-pin power plug.
- Failed Power Supply Unit (PSU) inside the desktop.
- Power outage or UPS battery fully drained during load shedding overnight.
- Loose internal components (RAM, GPU) causing a POST failure.

Troubleshooting Steps

1. Checked power cable connections at both the wall socket and PSU inlet — cable found slightly loose at the PSU end; reseated firmly.
2. Tested the wall socket using a phone charger — power confirmed present.
3. Inspected the power strip — switched on, no tripped indicator; reset button tested.
4. Confirmed APC Back-UPS Pro 1500VA was active with full charge — no load-shedding event detected.
5. Pressed power button after reseating the PSU cable — desktop powered on successfully; POST completed and Windows 11 Pro loaded.
6. Verified network connectivity post-boot: IP 192.168.20.12 confirmed via DHCP; gateway (192.168.1.1) reachable; NAS Finance share and Microsoft 365 accessible.
7. User confirmed Sage Business Cloud Accounting loaded with no data loss.

Root Cause

The power cable had worked itself partially free at the PSU inlet, likely due to movement of the workstation during the recent office setup. No hardware fault was identified.

Resolution

The power cable was reseated at the PSU inlet. The workstation powered on normally. The user was advised to avoid moving the desktop unit unnecessarily, and a cable tie was applied to secure the cable to the desk leg to prevent recurrence. No hardware replacement was required, and no data loss occurred.

Preventive Measures

- All desktop power cables to be checked and secured during the initial office setup phase.
- UPS units (APC 1500VA) confirmed operational — continue scheduled monthly UPS battery tests.
- Incident documented in helpdesk system; hardware inventory updated with status: No Fault Found.

4.2 INC-002 — No Internet Connectivity

Date Reported	20 May 2026
Reported By	Sales & Marketing Department (Multiple users, VLAN 20)
Assigned To	IT Support Specialist
Priority	High
Status	Resolved
Resolution Time	~40 minutes (including ISP contact)

Problem Description

Multiple users across the Sales and Marketing department (7 staff, workstations 192.168.20.18–.24) reported that they could not access the internet. Internal resources — including the Synology NAS (192.168.10.10) and network printers (192.168.30.10 and .11) — remained accessible, indicating the internal LAN was functioning correctly. The outage affected HubSpot CRM, Microsoft 365 (Outlook, Teams, OneDrive), Canva Pro, and general web browsing.

Symptoms

- Web browsers displayed "No Internet Connection" / "ERR_NAME_NOT_RESOLVED".
- Ping to external IP (8.8.8.8) failed; ping to internal devices succeeded.
- Internal NAS (192.168.10.10) and printers (192.168.30.10) remained accessible.
- Issue affected all devices across multiple departments, not just Sales.

- Microsoft Teams displayed a "No network connection" banner.

Possible Causes

- ISP outage or service interruption on the fibre line.
- Ethernet cable disconnected between the FortiGate router WAN port and the ISP modem/ONT.
- Router/firewall (FortiGate 60F) firmware crash or requiring reboot.
- DNS server failure — devices unable to resolve domain names.
- Firewall rule accidentally blocking all outbound traffic.
- ISP DHCP lease expired on the WAN interface.

Troubleshooting Steps

8. Confirmed scope across Finance and Software Development departments — internet down organisation-wide; internal LAN fully operational.
9. Accessed FortiGate 60F admin interface at 192.168.1.1 — WAN interface status showed "Disconnected"; confirmed upstream issue.
10. Inspected the Ethernet cable between the FortiGate WAN port and the ISP ONT — cable firmly connected at both ends; ONT link LED amber (no sync), indicating a line-level issue.
11. Powered off the ISP ONT; waited 60 seconds; powered back on — ONT link LED returned to green after approximately 90 seconds.
12. After ONT sync confirmed, performed a graceful reboot of the FortiGate 60F via the admin panel — router came back online in approximately 2 minutes.
13. Verified WAN interface showed "Connected" with a public IP assigned by the ISP; DNS servers confirmed as 8.8.8.8 / 8.8.4.4.
14. Tested internet connectivity: ping to 8.8.8.8 successful; Google, HubSpot CRM, and Microsoft 365 all accessible.
15. Sent an all-staff notification via Microsoft Teams confirming connectivity restored.
16. (Parallel action) Logged a support call with the ISP — ISP confirmed a brief line instability event in the area had been resolved; ISP reference number recorded.

Root Cause

ISP line instability caused the ONT to lose sync, dropping the WAN connection on the FortiGate router. The issue was upstream (ISP-side) and resolved after restarting the ONT and router. No internal network misconfiguration was found.

Resolution

The ISP ONT and FortiGate 60F router were restarted. WAN connectivity was restored within approximately 40 minutes of the ticket being raised. All internet-dependent services (HubSpot, Microsoft 365, Canva) were confirmed operational. DNS settings on the FortiGate were verified — Primary: 8.8.8.8, Secondary: 8.8.4.4 — no changes required.

Preventive Measures

- ISP SLA response time noted for future reference. Consider requesting a formal SLA document from the ISP guaranteeing minimum uptime (99.9%).
- Evaluate adding a secondary LTE/4G failover connection to the FortiGate for business continuity during future ISP outages.
- Schedule monthly router firmware checks via the FortiGate admin panel.
- Incident documented in helpdesk system with ISP reference number.

4.3 INC-003 — Network Printer Offline

Date Reported	21 May 2026
Reported By	Human Resources Department
Assigned To	IT Support Specialist
Priority	Medium
Status	Resolved
Resolution Time	~30 minutes

Problem Description

An HR department employee reported that the shared network printer (HP LaserJet Pro M404dn, Printer 1, VLAN 30, static IP: 192.168.30.10) was showing as "Offline" in Windows 11 Devices and Printers. Print jobs submitted by HR staff were queuing but not printing. The issue affected all users attempting to print to this device. Printer 2 (192.168.30.11) was unaffected. The HR department uses this printer for confidential documents including employee contracts, leave forms, and onboarding materials.

Symptoms

- Printer showed as "Offline" in Windows 11 Devices and Printers.
- Print queue showed jobs stuck with "Deleting" or "Error" status.
- Printer LCD panel showed no error codes; paper tray loaded.
- All users on the network unable to print to 192.168.30.10.
- Printer 2 (192.168.30.11) functioning normally.

Possible Causes

- Network cable disconnected from the printer or the managed switch port.
- Printer IP address changed (DHCP lease renewed with a different IP).
- Print Spooler service crashed or stopped on affected workstations.
- Printer powered off or entered deep sleep mode.
- Driver corruption or outdated print driver on user workstations.
- Printer memory full due to a large stuck print job.

Troubleshooting Steps

17. Physically inspected the printer — confirmed powered on; LCD panel showed "Ready"; paper tray loaded; no error lights or codes.
18. Inspected the Cat6 Ethernet cable connecting the printer to the managed switch (Cisco SG350-28) — cable firmly connected at both ends; switch port LED showed a solid green link light.
19. Printed a configuration page directly from the printer menu — current IP displayed as 192.168.30.15, indicating DHCP had assigned a new IP, overriding the previously configured address. Root cause identified.
20. Ran ping 192.168.30.15 from a workstation — successful; printer reachable at new IP.
21. Accessed the HP LaserJet Pro M404dn embedded web server at <http://192.168.30.15>; navigated to Network > IPv4 Settings; changed assignment from DHCP to Manual (Static); set IP: 192.168.30.10, Subnet: 255.255.255.0, Gateway: 192.168.1.1, DNS: 8.8.8.8; saved settings; printer rebooted automatically.
22. Updated the TCP/IP port on all affected HR workstations (3 machines) via Control Panel > Devices and Printers > Printer Properties > Ports tab to point to 192.168.30.10.
23. Restarted the Print Spooler service on each affected workstation via services.msc.
24. Cleared stuck print queues on each workstation by stopping Print Spooler, deleting all files in C:\Windows\System32\spool\PRINTERS, then restarting Print Spooler.
25. Removed the "Use Printer Offline" flag via Devices and Printers > Printer menu.
26. Submitted a test print from the HR workstation — job printed successfully; printer status showed "Ready".
27. Confirmed all 3 HR workstations could print successfully to 192.168.30.10.

Root Cause

The printer's IP address had been reassigned by DHCP from 192.168.30.10 to 192.168.30.15 following a lease renewal. The printer had not been configured with a static IP during initial setup, making it vulnerable to this issue. Windows workstations continued attempting to reach the printer at the old IP (192.168.30.10), causing the "Offline" status.

Resolution

A permanent static IP (192.168.30.10) was assigned to the printer via its embedded web server. The printer port was updated on all HR workstations, stuck print queues were cleared, and the Print Spooler service was restarted. Printing was restored for all HR users within 30 minutes. The IP Addressing Plan was updated to confirm Printer 1 as static at 192.168.30.10.

Preventive Measures

- Printer 2 (192.168.30.11) also checked and confirmed static IP assigned.
- All network devices (NAS, printers, access points, switch) to be configured with static IPs during initial setup to prevent DHCP-related outages.
- DHCP scope on the FortiGate to be reviewed: exclude static IP ranges (192.168.10.x, 192.168.30.x) from the DHCP pool to prevent conflicts.

- Incident documented in helpdesk system; IP Addressing Plan updated.

5. General IT Support Notes

The following operational standards apply to all IT support activity at Ubuntu Innovations (Pty) Ltd:

5.1 Ticketing & Escalation

- All incidents to be logged in the helpdesk ticketing system immediately upon receipt, with Ticket ID, date, reporter, priority, and status.
- Escalation path: IT Support Specialist > Senior Technician > IT Manager.

5.2 Critical Systems & Power

- Critical systems (NAS, router, switch) are protected by APC 1500VA UPS units.
- Monthly UPS battery tests to be scheduled.

5.3 Endpoint Security & Updates

- All devices run Windows 11 Pro with automatic updates enabled (scheduled 10:00 PM daily).
- Kaspersky Endpoint Security deployed with 24-hour definition updates on all endpoints.

5.4 Data Security

- Finance and HR data stored on the NAS (192.168.10.10) in encrypted, access-controlled folders.
- Any incident involving Finance or HR data folders must be escalated immediately and logged as a potential data security incident.

5.5 Remote Access

- Remote access incidents (Executive Management and authorised staff) to be handled via the FortiGate VPN.
- Check VPN tunnel status in the FortiGate admin panel at 192.168.1.1.

6. Conclusion

All three incidents logged during Week 1 were resolved successfully within 40 minutes, with no data loss and minimal business disruption. The incidents highlighted the importance of securing hardware connections during office setup (INC-001), establishing ISP SLA agreements and

failover connectivity (INC-002), and ensuring all network devices are configured with static IP addresses from initial deployment (INC-003).

Preventive measures have been recommended for each incident and will be tracked through the helpdesk system to ensure implementation. These actions are expected to reduce the likelihood of similar incidents occurring in subsequent weeks.

End of Report — Week 1 | Ubuntu Innovations (Pty) Ltd | Capaciti Tech Career Accelerator | May 2026